

Sandeep Anand

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Skills

Languages: Python, C++, SQL

ML/AI: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Agentic AI, AI Agents, Prompt Engineering, LLM Evaluation

Libraries/Frameworks: TensorFlow, Keras, LangChain

Backend & APIs: FastAPI, Flask

MLOps & DevOps: Docker, CI/CD, Jenkins, Ansible, Git, Rundeck

Cloud : Azure

Databases & Storage: MySQL, SQLite, Vector Databases (FAISS, ChromaDB, Pinecone), Semantic Search, Embeddings

Tools: Jupyter, VS Code, Streamlit

Work Experience

Halocom, Hyderabad

May 2025 - Jan 2026

AIML Engineer

- **Production AI Systems:** Designed and deployed real-time ML-driven conversational systems integrating ASR, NLP, LLMs, TTS, and decision logic for large-scale user interactions.
- **Latency Optimization:** Reduced end-to-end inference latency by **40%**, achieving **2s** response time through pipeline and API optimization.
- **Chatbot & RAG:** Developed voice chatbot modules using ASR–LLM–TTS pipeline, integrated RAG-based retrieval, vector databases, and LangChain-based orchestration for domain-specific responses.
- **Agentic Workflows:** Built agentic AI decision pipelines for intent handling, contextual routing, and response generation in conversational systems.
- **Deployment:** Deployed the entire pipeline using CI/CD, enabling automated updates and reliable production rollout.
- Awarded **Star Performer** for high-impact contributions to production AI systems.

HSBC, Pune

January 2025 - March 2025

Software Engineer

Custom Job Scheduler for Ansible

- Designed and implemented a scalable job scheduling system supporting automated data and ML-related workflows with authentication, execution tracking, and audit logs.
- Built reliable backend services to orchestrate task execution, retries, and failure handling across distributed environments.
- Integrated the scheduler with Ansible and Jenkins to enable CI/CD-driven pipeline execution.
- Successfully replaced Rundeck, improving system reliability and standardizing deployments across Dev, UAT, and Production.

Team Infits, Hyderabad

May 2024 - October 2024

Machine Learning Intern

- Developed NLP-driven systems for text processing, content analysis, and automation, focusing on scalable and production-ready ML solutions.
- Built a Devanagari text conversion pipeline to transform legacy Akruti-encoded text into Unicode format, enabling downstream NLP and analytics tasks.
- Designed an NLP-based LinkedIn profile analysis system to evaluate profile quality and generate role-specific optimization recommendations.
- Worked on LLM-based question generation and validation pipelines using GSM8K-style reasoning tasks, leveraging Llama models and similarity-based evaluation.
- Automated notification workflows to integrate ML outputs into user-facing systems, improving information delivery efficiency.
- Worked on LLM evaluation, alignment, and RLHF-style feedback pipelines by generating, validating, and scoring LLM outputs for reasoning quality, factual accuracy, and semantic similarity.

Data Science

- **Implemented advanced ML techniques:** Focused on Few-shot learning, Zero-shot learning, and CLIP architecture.
- **Curated and analyzed datasets:** Worked with Trademark Character Dataset (70 characters) and Car Logo Dataset (13 logos).
- **Achieved remarkable results:** Conducted comprehensive comparisons with state-of-the-art models, leading to impressive outcomes.

Education

IIIT Kurnool

B.Tech in Computer Science and Engineering

Dec 2021 - April 2025

CGPA: 8.00/10

Relevant Coursework: Object Oriented Programming, Data Structures and Algorithms, Operating Systems, Database Management System

Project Work

- **Exploring Advanced DL Architectures for MURA Dataset (2025):** Experimented with state-of-the-art models, including Fourier Transforms and Triplet Networks, to improve musculoskeletal radiograph detection. Evaluated model performance using Cohen's Kappa and conducted comparative studies to enhance diagnostic accuracy. Python, TensorFlow. 
- **Voice Authentication-Based Digital Payment App for Indian Regional Languages (2024):** Developed a voice-based digital payment app with authentication support for regional Indian languages. Implemented a customized voice authentication system and improved model accuracy through iterative enhancements and extensive testing. Python, SVM. 
- **Smart LLM Router & Cost-Quality Optimisation System (2025):** Designed a domain-aware agentic **LLM routing system** using historical prompt replay to optimise cost, quality, and latency through intelligent model selection. Benchmarked 6+ LLMs across multiple domains and complexity levels, and implemented an **LLM-as-a-Judge** to convert subjective response quality into quantitative scores. Built deterministic, preference-based scoring logic (cost / quality / latency) and exposed the system via Flask APIs with a Streamlit dashboard for transparent model comparison. Containerized and deployed using Docker for scalable model benchmarking. *Python, LLMs, Agentic AI, LangChain, Flask, Streamlit, Pandas, Prompt Engineering*


Achievements

- **Winner, HSBC Hackathon 2024:** Secured a 3-month internship opportunity by winning the final round hackathon.
- **Mentor, HSBC Hackathon 2025:** Guided participants through solution design and implementation.
- **Finalist, Blend 360 Hackathon & Portkey AI Hackathon:** Reached the final round among top-performing teams.
- Selected for **Samsung Innovation Campus** Internship.
- **Coordinator of DataWorks:** Organized various sessions on Machine Learning and Deep Learning for 200+ students.
- Solved 700+ Data Structures and Algorithm questions on platforms like GFG, Leetcode, and CodeChef.
- Led hackathons, coding contests, and expert talks as **CSE Technical Lead** during the institute fest.
- **NSS Core Team Member:** Organized blood donation camps, tree plantation drives, and college cleaning drives.